

SignMuon, MuonSign, and the Role of Error Feedback

Anonymous submission

Abstract

SignMuon — sign compression of the Muon update to one bit per parameter — is the most straightforward adaptation of Muon to an extremely low communication budget. It outperforms SignSGD in practice, yet it can diverge on a linear function. The natural conjecture is that the order of operations is to blame, and that signing before rather than after the Linear Minimization Oracle (LMO) would fix it. It does not: MuonUSign diverges too, as does MuonSign, which signs both sides, so no naive placement of the sign around the LMO converges. Error feedback repairs one of them — EF21-MuonUSign attains the standard $\mathcal{O}(T^{-1/2})$ rate for smooth nonconvex optimization and decisively outperforms SignSGD, and compressing the downlink as well costs little beyond that. What the guarantee costs is the placement. Across centralized, federated and language-modelling experiments the best placement is consistently sign-*after*-the-LMO — precisely the one our counterexamples break and error feedback fails to repair — while the provably convergent sign-*before* variants trail it by about two accuracy points in federation. We report this tension rather than resolve it: the counterexamples constrain worst-case instances, not CIFAR-10.

1 Introduction

Training a deep network across clients costs bandwidth as well as computation: every round, each client must ship a full update, and under a limited link those updates dominate the wall-clock cost. Compression is the standard remedy (Bernstein et al. 2018, 2019; Beznosikov et al. 2024), and SignSGD is its extreme point — one bit per coordinate, at little cost in final accuracy. But SignSGD flattens each weight matrix into a vector, discarding the two-dimensional structure that matrix-aware optimizers exploit. Muon exploits precisely that structure: it orthonormalizes the momentum matrix before stepping, which stabilizes training and, in several settings, beats tuned adaptive methods (Jordan et al. 2024; Bernstein and Newhouse 2024; Essential AI 2025). What it sends, however, is a *dense* matrix at full precision — the one thing a bandwidth-limited system cannot afford. Matrix geometry and a one-bit budget pull in opposite directions.

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We study the natural ways to combine sign compression with the Muon LMO at one bit per parameter: **SignMuon**, which signs *after* the LMO, $\text{sign}(\text{LMO}(\cdot))$; **MuonUSign**, which signs *before*, $\text{LMO}(\text{sign}(\cdot))$; and **MuonSign**, which signs on *both* sides and so compresses uplink and downlink alike. All three keep Muon’s matrix-aware geometry, but only sign-*after* reaches Muon’s accuracy: at $N = 11$ clients the three span five accuracy points, in the order after, before, both sides, the last of them falling below SignSGD.

None of them, however, has a global convergence guarantee. We prove that all three can diverge — SignMuon on a 4×4 instance (Theorem 1), MuonUSign and MuonSign on a single common 5×5 one (Theorems 2–3) — the 1-bit step becoming an ascent direction on a linear objective. To restore convergence at the same communication volume we introduce **EF21-MuonUSign**, adapted from EF21-Muon (Grunkowska et al. 2025): the scaled sign compresses only the residual between the momentum and a gradient estimator, and the descent step takes the full Muon LMO of the reconstructed estimator. **EF21-MuonSign** adds a second error-feedback loop on the downlink, compressing both channels to one bit. Both descend on the counterexamples above.

We evaluate the methods on centralized CIFAR-10 with a ResNet-18, federated CIFAR-10 with a CNN at $N = 11$ clients, the nanoGPT speedrun, and a synthetic smooth convex least-squares problem. Our contributions are: (i) SignMuon, MuonUSign and MuonSign, matrix-aware 1-bit optimizers placing the sign after, before and on both sides of the LMO; (ii) counterexamples proving all three may diverge (Theorems 1–3), at essentially minimal sizes; (iii) a universal divergence theorem (Theorem 4): error feedback on the LMO *output* repairs SignMuon for no (L, η, μ) ; (iv) EF21-MuonUSign, restoring the guarantee at a one-bit uplink, and (v) EF21-MuonSign, adding a downlink loop to keep both channels at one bit; and (vi) a tuned five-seed federated comparison of all six at one budget, separating the three placements single-seed evidence had shown as coincident and locating the guarantee’s cost at the placement, not the downlink. Proofs, algorithms, and the numbered results cited but not stated below are in the supplementary material.

2 Related Work

SignSGD. Vanilla SignSGD, introduced by Bernstein et al. (2018) and later applied to federated learning by Bernstein

et al. (2019), required increasing batch sizes to guarantee convergence. This issue, together with SignSGD’s inherent bias, was addressed through the adoption of the Error Feedback technique (Karimireddy et al. 2019). An alternative approach is to randomize the sign operator itself (Chen et al. 2020; Safaryan and Richtarik 2021; Jin et al. 2025).

Compression for Muon. Gruntkowska et al. (2025) developed a theoretical framework for federated learning with error feedback in which Muon/Gluon (Riabini et al. 2025) updates are compressed bidirectionally (client-to-server and server-to-client). Qian et al. (2026) employed variance reduction to lower the compression error for Gluon under unbiased and contractive compressors. Concurrently with our work, Mishra, Trivedi, and Kumar (2026) proposed vanilla SignMuon and stated an $\mathcal{O}(1/\sqrt{T})$ guarantee for it; in Appendix A.6 we show that their finalized rate covers only the gradient-sign instantiation, leaving the bound for the actual polar-sign method uninformative on exactly the instances of Theorem 1. Bolatov et al. (2026) *alternate* spectral (Muon) and sign (Lion-type) steps rather than composing the sign with the LMO, so the method they too call SignMuon is not the one studied here and our counterexamples do not bear on it.

Changing the norm instead of compressing the oracle. Kravatskiy et al. (2025), working with the Ky Fan norms, mix sign and spectral geometry *inside* the oracle, obtaining **S-Muon**, which is therefore still an LMO for an explicit norm and keeps the convergence theory of one. Despite the name it is not one of the methods studied here: our three placements put the sign *around* the oracle, where it is a compressor and not a geometry (Appendix A.7).

3 Problem Statement

We consider the stochastic optimization problem

$$\min_{\mathbf{X} \in \mathcal{X}} \{f(\mathbf{X}) := \mathbb{E}_{\xi \sim \mathcal{D}}[f_\xi(\mathbf{X})]\}, \quad (1)$$

where $\mathcal{X}$ is the parameter space (e.g., $\mathbb{R}^d$ or $\mathbb{R}^{m \times n}$) and the $f_\xi : \mathcal{X} \rightarrow \mathbb{R}$ are continuously differentiable and possibly non-convex: $f_\xi(\mathbf{X})$ is the loss of a model $\mathbf{X}$ at a data point $\xi \sim \mathcal{D}$. In the federated setting the data are split across clients, and the training problem becomes

$$\min_{\mathbf{X} \in \mathcal{X}} \left\{ f(\mathbf{X}) := \frac{1}{N} \sum_{j=1}^N f_j(\mathbf{X}) \right\}, \quad f_j(\mathbf{X}) = \mathbb{E}_{\xi_j \sim \mathcal{D}_j} [f_j(\mathbf{X}; \xi_j)] \quad (2)$$

where $N \geq 1$ is the number of clients and f_j is the loss on the data $\mathcal{D}_j$ held by client $j \in [N] := \{1, \dots, N\}$ (Bernstein et al. 2019; Horváth et al. 2019; Beznosikov et al. 2024; Gruntkowska et al. 2025; Ahn et al. 2025).

Two assumptions run through the paper. The model is a tuple of matrix layers $\mathbf{X} = [\mathbf{X}_1, \dots, \mathbf{X}_p]$, $\mathbf{X}_i \in \mathbb{R}^{m_i \times n_i}$, with ∇_i the gradient block of layer i and $\|\nabla f(\mathbf{X})\|_*^2 := \sum_i \|\nabla_i f(\mathbf{X})\|_*^2$; a single matrix parameter is the case $p = 1$.

Assumption 1 (Lower boundedness) $f(\mathbf{X}) \geq f^*$ for all $\mathbf{X}$; where explicitly invoked, each local $f_j \geq f_j^*$ as well.

Assumption 2 (Layer-wise smoothness) Each layer carries the spectral norm $\|\cdot\|_{2 \rightarrow 2}$, whose dual is the nuclear norm $\|\cdot\|_*$. For every layer i and all $\mathbf{X}, \mathbf{Y}$,

$$\|\nabla_i g(\mathbf{X}) - \nabla_i g(\mathbf{Y})\|_* \leq L_i^g \|\mathbf{X}_i - \mathbf{Y}_i\|_{2 \rightarrow 2} \quad (3)$$

for $g = f$ (constant L_i) and $g = f_j$ (constant $L_{i,j}$). A weaker layer-wise (L^0, L^1) form, stated in Appendix A.1, is used only in Corollary 2.

A Linear Minimization Oracle (LMO) minimizes a linear form over the unit ball of a norm, returning the steepest-descent direction in that norm (Appendix A.1). Muon takes that norm to be spectral on each matrix layer, which turns the oracle into an orthogonalization: writing $\mathbf{M} = \mathbf{U}\Sigma\mathbf{V}^\top$ for the rank- r singular value decomposition of $\mathbf{M} \in \mathbb{R}^{m \times n}$, the LMO direction is

$$A(\mathbf{M}) = -\mathbf{U}\mathbf{V}^\top \in \mathbb{R}^{m \times n}. \quad (4)$$

In practice Muon applies the oracle to a smoothed gradient estimate with momentum:

$$\begin{aligned} \mathbf{M}_t &= \mu \mathbf{M}_{t-1} + \mathbf{G}_t, \\ \mathbf{M}_t &= \mathbf{U}_t \Sigma_t \mathbf{V}_t^\top, \\ \mathbf{D}_t &= -A(\mathbf{M}_t) = \mathbf{U}_t \mathbf{V}_t^\top \\ \mathbf{X}_{t+1} &= \mathbf{X}_t - \eta_t \mathbf{D}_t. \end{aligned} \quad (5)$$

Muon requires transmitting the full update matrix $\mathbf{D}_t$, which is expensive in federated settings. Our objective is to combine its matrix-aware geometry with extreme (1-bit) communication compression. SignMuon, MuonUSign, and MuonSign achieve it empirically, but lack global convergence guarantees (Theorems 1–3). The Error-Feedback variants restore convergence: in Appendix A.9 we show that EF21-MuonUSign and EF21-MuonSign are *exact instances* of the EF21-Muon framework of Gruntkowska et al. (2025), and so inherit the standard $\mathcal{O}(T^{-1/2})$ rate for smooth non-convex optimization (Theorem 5):

$$\min_{0 \leq t \leq T} \mathbb{E}[\|\nabla f(\mathbf{X}_t)\|_*^2] = \mathcal{O}(T^{-1/2}), \quad (6)$$

as well as an $\mathcal{O}(T^{-1/4})$ rate for $\min_t \sum_i \mathbb{E} \|\nabla_i f(\mathbf{X}_t)\|_*$ under (L^0, L^1) -smoothness with a plain constant learning rate (Corollary 2). Both displays assume one step-size constant common to all layers; Corollaries 1–2 give the general weighted form. Finally, EF21-MuonSign adds a second error-feedback loop on the downlink, making communication 1-bit in *both* directions. The $\mathcal{O}(T^{-1/2})$ rate survives, but not for free: a sign-based downlink is contractive only in the Euclidean norm, whereas Muon’s spectral geometry asks for contractivity in the layer norm, so the admissible step size shrinks by a further $\sqrt{r}$ in the layer rank. The gap is not an artefact of the analysis — the scaled sign is spectrally contractive for *no* parameter at all, so only a different compressor helps (Remark 4) — and we measure what it costs on a transformer (Section 5.3).

4 Theory

Consider the **SignA** family, where A is any optimizer built on LMO directions. A SignA method initializes $\mathbf{M}_0 = \mathbf{0}$ and at each iteration $t \geq 1$ performs:

$$\begin{aligned} \mathbf{M}_t &= \mu \mathbf{M}_{t-1} + \mathbf{G}_t && \text{(Momentum accumulation)} \\ \tilde{\mathbf{M}}_t &= \begin{cases} \mathbf{G}_t + \mu \mathbf{M}_t & \text{(Nesterov momentum),} \\ \mathbf{M}_t & \text{(otherwise)} \end{cases} && \text{(Effective direction),} \\ \mathbf{D}_t &= -A(\tilde{\mathbf{M}}_t), && \text{(LMO direction)} \\ \mathbf{s}_t &= \text{sign}(\mathbf{D}_t) \in \{\pm 1\}^{m \times n}, && \text{(Sign compression)} \\ \mathbf{X}_{t+1} &= \mathbf{X}_t - \eta_t \mathbf{s}_t && \text{(Parameter update),} \end{aligned} \quad (7)$$

where $\mathbf{X}_t \in \mathbb{R}^{m \times n}$ is the parameter matrix, $\mathbf{G}_t$ its stochastic gradient, $\mu \in [0, 1)$ the momentum coefficient and η_t the learning rate; $A(\cdot)$ is a Linear Minimization Oracle (LMO) in a specified non-Euclidean norm, and $\text{sign}(\cdot)$ acts element-wise on the structured direction $\mathbf{D}_t$.

Exact zeros. We resolve $\text{sign}(0)$ to an independent random ± 1 rather than to 0, on every channel in this paper: a third symbol would make the alphabet ternary and the channel wider than one bit, whereas randomizing costs nothing in expected descent and makes exact two statements the ternary convention only approximates — $\|\mathbf{s}_t\|_F = \sqrt{mn}$, which the step-size rule below assumes, and the contraction identity of Lemma 4 (Appendix A.9). Everything below is stated under this convention, but none of the counterexamples invokes it: no instance in Theorems 1–4 has a zero entry.

SignMuon is a particular instance: it combines the orthogonalized LMO directions of Muon with the sign compression of SignSGD. MuonUSign is its mirror image, applying the sign *before* the LMO — it compresses the effective direction to $\text{sign}(\tilde{\mathbf{M}}_t)$ and steps along the Muon LMO of that sign matrix (Algorithm A4):

$$\mathbf{D}_t = -A(\text{sign}(\tilde{\mathbf{M}}_t)), \quad \mathbf{X}_{t+1} = \mathbf{X}_t - \eta_t \mathbf{D}_t. \quad (8)$$

MuonSign (Algorithm A5) signs both sides, additionally compressing the LMO output to $\text{sign}(\mathbf{D}_t)$, so that both the uplink and the downlink cost one bit per parameter.

4.1 Centralized Learning

In the centralized setting, SignMuon is applied to the matrix-valued parameters (weights of convolutional and fully connected layers, $n_{\text{dim}} \geq 2$), while vector parameters (biases, BatchNorm, $n_{\text{dim}} = 1$) and the final classification layer are optimized with AdamW — the design introduced by Jordan et al. (2024) and formalized in (Li and Hong 2025; Riabinin et al. 2025).

For matrix parameters the key component is the orthonormal LMO direction $\mathbf{D}_t \approx \mathbf{U}_t \mathbf{V}_t^\top$, which we approximate by a 5th-order Newton–Schulz iteration rather than a full SVD at every step (Algorithm A1). SignMuon signs that direction, so it transmits one bit per component while keeping the geometry of orthonormal updates (Algorithm A2 in Appendix A.18); MuonSign is built from the same idea (Algorithm A5). One further implementation choice is not free, and we state it here because every experiment depends on it.

Per-layer step sizes: the unit-gain heuristic. The methods of this paper produce step matrices from two families with *exactly* known Frobenius norms: $\|\mathbf{UV}^\top\|_F = \sqrt{r}$ for the LMO-terminated methods ($r = \min(m, n)$), and $\|\mathbf{s}\|_F = \sqrt{mn}$ for the sign-terminated ones. These scale differently with layer shape, so no single global η is right for both families, or across layers within one. We fix the shape dependence a priori and tune only a shape-free base rate η_0 , setting $\eta_\ell = \eta_0 \lambda_\ell$ with

$$\lambda_\ell = \frac{\sqrt{m}}{\|\mathbf{s}_\ell\|_F}, \quad \text{i.e.} \quad \lambda_\ell^{\text{LMO}} = \sqrt{\max(1, \frac{m}{n})}, \quad \lambda_\ell^{\text{SIGN}} = \frac{1}{\sqrt{\text{fan-in}}}. \quad (9)$$

The criterion is that every layer’s update have the same root-mean-square input–output gain, the average-case form of the spectral scaling condition (Yang, Simon, and Bernstein 2023; Large et al. 2024); η_0 is then literally the per-step RMS gain. We treat (9) as a heuristic and use it uniformly, for one reason above all: its LMO branch *reproduces* the aspect-ratio factor $\sqrt{\max(1, m/n)}$ that the reference Muon implementation already ships as a practical fix (Jordan et al. 2024), so a criterion already accepted in practice for one family supplies the missing counterpart, $1/\sqrt{\text{fan-in}}$, for the other. Appendix A.17 gives the derivation, the measurement that selects the exponent $\frac{1}{2}$ over μP ’s 1, the caveats, and the sensitivity of our conclusions to the choice.

4.2 Theoretical limitations

A natural hope is that the Muon LMO direction — the steepest-descent direction in the spectral norm — survives sign compression, whether the sign is applied before or after the oracle. It does not: in each of the three placements the compressed step can become an *ascent* direction on a smooth objective.

Write $\text{polar}(\mathbf{Y}) := -A(\mathbf{Y}) = \mathbf{UV}^\top$ for the Muon LMO direction, read off the rank-truncated SVD $\mathbf{Y} = \mathbf{U}\Sigma\mathbf{V}^\top$. At rank-deficient $\mathbf{Y}$ the spectral-ball LMO is not unique, and this selection — the one our implementation makes — is part of what the counterexamples assert. The three methods share the update $\mathbf{X}_{t+1} = \mathbf{X}_t - \eta_t \mathbf{s}_t$ and differ only in where the sign acts on the effective direction $\tilde{\mathbf{M}}_t$:

$$\mathbf{s}_t = \begin{cases} \text{sign}(\text{polar}(\tilde{\mathbf{M}}_t)) & \text{(SignMuon: after),} \\ \text{polar}(\text{sign}(\tilde{\mathbf{M}}_t)) & \text{(MuonUSign: before),} \\ \text{sign}(\text{polar}(\text{sign}(\tilde{\mathbf{M}}_t))) & \text{(MuonSign: both).} \end{cases} \quad (10)$$

All three transmit one bit per parameter on the uplink. On the downlink SignMuon and MuonSign distribute a ± 1 -valued object and so cost one bit in each direction, while MuonUSign’s server-side $\text{polar}(\cdot)$ is dense and costs full precision.

We refute the descent property on *linear* objectives—the simplest smooth functions, on which any sound method must make unbounded progress:

$$f(\mathbf{W}) = \langle \mathbf{G}, \mathbf{W} \rangle = \text{Tr}(\mathbf{G}^\top \mathbf{W}), \quad \nabla f(\mathbf{W}) \equiv \mathbf{G}. \quad (11)$$

Gradient descent and full-precision Muon drive $f \rightarrow -\infty$ here; a rule that instead drives $f \rightarrow +\infty$ is unambiguously

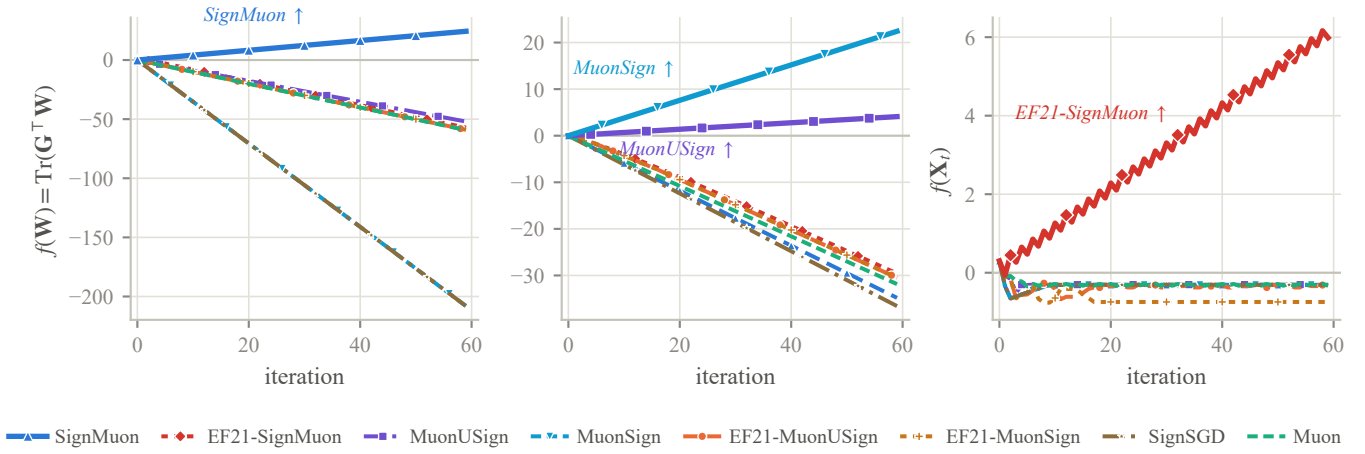


Figure 1: All eight methods on the three counterexample instances; in each panel the ascending method is drawn heavy and named. *Left*: SignMuon on the 4×4 instance of Theorem 1. *Centre*: MuonUSign and MuonSign on the shared 5×5 instance of Theorems 2–3; both panels plot the linear objective (11). *Right*: EF21-SignMuon on the 2×2 instance of Theorem 4, in normalized units with $\eta = 1$. Trajectories are momentum-free: for the three placements Proposition 1 shows this is without loss, and for EF21-SignMuon Lemma 3 supplies a separate instance per $(\mu, \text{variant})$ generating the same target sequence; the remaining baselines are shown at $\mu = 0$ only.

ascending. Such an f is not bounded below, so Assumption 1 fails — but only formally: the iterates travel a fixed ray on which f increases, and flattening f far below that ray restores the assumption without moving a single trajectory of this section — the device used again, in detail, for Theorem 4 (Remark 1).

Momentum plays no role on (11). The buffer is $\tilde{\mathbf{M}}_t = \gamma_t \mathbf{G}$ with $\gamma_t > 0$ under both the Standard and the Nesterov rule, and $\text{sign}(\cdot)$ and $\text{polar}(\cdot)$ are invariant under positive scaling, so the step is the constant matrix $\mathbf{s}(\mathbf{G})$ and

$$f(\mathbf{X}_{t+1}) - f(\mathbf{X}_t) = -\eta_t \langle \mathbf{G}, \mathbf{s}(\mathbf{G}) \rangle \quad (12)$$

for every $\mu \in [0, 1)$ and either variant (Proposition 1, Appendix A.2). One gradient with $\langle \mathbf{G}, \mathbf{s}(\mathbf{G}) \rangle < 0$ thus makes f increase at *every* step, at *any* step size, with $f(\mathbf{X}_t) \rightarrow +\infty$ whenever $\sum_t \eta_t = \infty$.

We exhibit such a gradient for each placement. For SignMuon it is a $\mathbf{G} \in \mathbb{R}^{4 \times 4}$ built as $1000 \mathbf{u}_1 \mathbf{v}_1^T + \mathbf{O}$ with $\mathbf{O}$ orthogonal, so that $\text{polar}(\mathbf{G}) = \mathbf{O}$ while the sign pattern of $\mathbf{O}$ is anticorrelated with the dominant component, giving $\langle \mathbf{G}, \mathbf{s}(\mathbf{G}) \rangle = -\frac{42468}{103}$. Signing *before* the oracle fares no better — the spectral LMO can rotate a sign matrix away from $\mathbf{G}$ rather than heal it — and a single $\mathbf{G} \in \mathbb{R}^{5 \times 5}$ defeats both remaining placements at once, at $\langle \mathbf{G}, \mathbf{s}(\mathbf{G}) \rangle \approx -13.89$ for MuonUSign and -76 for MuonSign. These are Theorems 1–3, with the constructions in Appendices A.3–A.5; the left and centre panels of Figure 1 show the resulting ascents.

Both sizes are essentially minimal. SignMuon provably cannot ascend at 2×2 , and no 3×3 instance was found; MuonUSign and MuonSign provably cannot ascend at 4×4 or smaller, by exhaustive enumeration of sign patterns (Appendices A.3–A.4). No placement of the sign around the Muon LMO, then, yields a convergent method.

4.3 Centralized Error Feedback: EF21-SignMuon

Theorems 1–3 rule out every static placement of the sign around the LMO. The classical remedy for a biased one-bit compressor is *error feedback* — store the compression error and reinject it at the next step — which repairs SignSGD (Karimireddy et al. 2019) and was later sharpened into EF21 (Richtárik, Sokolov, and Fatkhullin 2021). Its most direct use for SignMuon keeps the geometry — the sign *after* the LMO — and error-feedbacks the oracle’s output. The resulting method, **EF21-SignMuon** (Algorithm A3), tracks an estimate $\mathbf{d}_t^{\text{est}}$ of the LMO direction $\mathbf{D}_t = \text{polar}(\tilde{\mathbf{M}}_t)$, refreshed by a scaled sign of the residual,

$$\begin{aligned} \mathbf{d}_t^{\text{est}} &= \mathbf{d}_{t-1}^{\text{est}} + \alpha_t \text{sign}(\mathbf{D}_t - \mathbf{d}_{t-1}^{\text{est}}), \\ \alpha_t &= \text{mean}|\mathbf{D}_t - \mathbf{d}_{t-1}^{\text{est}}|, \end{aligned} \quad (13)$$

and steps $\mathbf{X}_t = \mathbf{X}_{t-1} - \eta_t \mathbf{d}_t^{\text{est}}$.

Error feedback on the compressed *output*, however, does not restore the descent property—and it fails *universally*, at every problem scale and momentum setting.

Theorem 4 (Divergence of EF21-SignMuon) *For every $L > 0$, step size $\eta > 0$, momentum coefficient $\mu \in [0, 1)$, and either momentum variant, there is an L -smooth (Assumption 2), bounded-below (Assumption 1) function $f : \mathbb{R}^{2 \times 2} \rightarrow \mathbb{R}$ on which EF21-SignMuon started at $\mathbf{X}_0 = \mathbf{0}$ diverges: for an explicit constant $c = c(f) > 0$,*

$$f(\mathbf{X}_{t+2}) - f(\mathbf{X}_t) = cL\eta^2 > 0 \quad (t \geq 3), \quad (14)$$

so $f(\mathbf{X}_t) \rightarrow +\infty$. In particular, no step-size rule $\eta = \eta(L, \mu)$ using only the smoothness and momentum constants can make the method convergent.

The obstruction (proof in Appendix A.8) is specific to signing the LMO output: a *single* magnitude α_t rescales all entries’ signs at once, so a large, sign-flipping off-diagonal pins

$\alpha_t = \Theta(1)$ and swamps the small diagonal target, whose estimate then locks onto the *wrong* sign and drives one coordinate of $\mathbf{X}_t$ to infinity. This is not a degeneracy—the matrices driving the divergent tail have condition number $\frac{16}{9}$ —and momentum does not help: EF21-SignMuon is the lone ascending method in the right panel of Figure 1, and Figure A1 (Appendix A.8) records the same rate $\frac{49}{480}$ per step for every μ and both momentum variants. The remedy is to error-feedback the gradient *before* the LMO rather than its output—the subject of the next subsection.

4.4 Centralized Error Feedback: EF21-MuonUSign and EF21-MuonSign

Theorem 4 pinpoints the defect of EF21-SignMuon, and it is not the shared magnitude α_t as such: the methods below couple all coordinates through the very same scalar. It is *what the estimator is asked to track*. Error feedback needs a target that settles — the EF21 recursion contracts only because the target moves $\mathcal{O}(\eta)$ per round — whereas the LMO output $\text{polar}(\tilde{\mathbf{M}}_t)$ is a non-Lipschitz function of the momentum and can jump by $\Theta(1)$ between consecutive rounds however small η is, which is exactly what the 2×2 instance arranges. Compressing the *gradient* instead leaves the estimator chasing a target that moves with the step size, and, as we show next, restores convergence at the same one-bit uplink cost. Concretely, we replace the sign-compressed step $\mathbf{X}_{t+1} = \mathbf{X}_t - \eta_t \text{sign}(\mathbf{D}_t)$ by an error-feedback variant that adapts the EF21-Muon mechanism (Gruntkowska et al. 2025) to sign compression.

The **USign** compressor is the directional pair $(\mathcal{C}_\uparrow, \mathcal{C}_\downarrow)$ with $\mathcal{C}_\uparrow(\mathbf{Y}) = \text{mean}|\mathbf{Y}| \text{sign}(\mathbf{Y})$ the scaled sign on the uplink and $\mathcal{C}_\downarrow = I$ the identity on the downlink: one bit per parameter from client to server, a full-precision model back. **EF21-MuonUSign** pairs it with EF21 and the Muon LMO.

EF21-MuonUSign maintains a gradient estimator $\mathbf{g}_t^{\text{est}}$ updated via a contractive sign compressor on the residual $\Delta_t = \mathbf{M}_t - \mathbf{g}_t^{\text{est}}$:

$$\alpha_t = \text{mean}(|\Delta_t|), \quad \mathbf{g}_{t+1}^{\text{est}} = \mathbf{g}_t^{\text{est}} + \alpha_t \text{sign}(\Delta_t), \quad (15)$$

and takes the descent step

$$\mathbf{X}_{t+1} = \mathbf{X}_t - \eta_t \mathbf{D}_t, \quad \mathbf{D}_t = -A(\mathbf{g}_{t+1}^{\text{est}}) \approx \mathbf{U}_{t+1} \mathbf{V}_{t+1}^\top. \quad (16)$$

Here $\mathbf{U}_{t+1} \mathbf{V}_{t+1}^\top$ is the polar factor of the estimator, not of ∇f , so no step-wise descent property is claimed; what it buys is an estimator tracking a target that moves at the step size — the hypothesis the transferred guarantee needs (Corollary 1). The sign in (15) acts only on the internal compression residual, so the uplink stays at one bit per parameter (Algorithm A6, Appendix A.18). Figure 1 confirms that EF21-MuonUSign descends on the counterexample on which SignMuon ascends.

EF21-MuonSign makes both channels one-bit by applying the same scaled-sign operator to the downlink. The mechanism is a second error-feedback loop, on the *model* rather

than the gradient, and it splits the method into two iterates:

$$\begin{aligned} \mathbf{X}_{t+1} &= \mathbf{X}_t - \eta_t \mathbf{D}_t && \text{(server model),} \\ \mathbf{W}_{t+1} &= \mathbf{W}_t + \alpha_t^\downarrow \text{sign}(\mathbf{X}_{t+1} - \mathbf{W}_t) && \text{(broadcast model),} \end{aligned} \quad (17)$$

with $\mathbf{D}_t = -A(\mathbf{g}_{t+1}^{\text{est}})$ as before. The server model $\mathbf{X}$ takes the exact LMO step and never leaves the server; what crosses the downlink is the one-bit refresh of the broadcast model $\mathbf{W}$, and $\mathbf{W}$ is the only model the rest of the method ever sees — gradients, momentum and the uplink residual are all computed there. Training happens at $\mathbf{W}$; the transferred guarantee (Corollary 1) bounds the gradient at $\mathbf{X}$. Under an exact downlink the two coincide and the method reduces to EF21-MuonUSign; how far they drift apart under the one-bit downlink is measured in Section 5.3. The full procedure is Algorithm A7; centrally the “channels” are internal to the update rule, which simply maintains both matrices.

Both error-feedback methods pay for the guarantee in memory: one buffer per compressed channel (the gradient estimate $\mathbf{g}^{\text{est}}$; for EF21-MuonSign also the broadcast model $\mathbf{W}$). The overhead is small on the server but may be unwanted on memory-limited workers.

4.5 Federated Learning

In the federated setting (2) each client accumulates a stochastic gradient over n_{steps} local mini-batches, maintains a local momentum buffer, applies the Muon LMO (Newton–Schulz, Algorithm A1), and uploads the elementwise sign of the result; the server takes a majority vote, $\mathbf{s}_t^{\text{agg}} = \text{sign}(\sum_{j=1}^N \mathbf{s}_t^{(j)})$, and the step is $\mathbf{X}_{t+1} = \mathbf{X}_t - \eta_t \mathbf{s}_t^{\text{agg}}$. The uplink costs one bit per parameter, and so does the *downlink*: $\mathbf{s}_t^{\text{agg}}$ is ± 1 -valued in every coordinate — exactly, at an odd client count, under the convention of Section 4 — so the server broadcasts the vote rather than the model and each client applies it to its own replica. Replicas start from a common $\mathbf{X}_0$ and receive identical updates, so they never drift. This is a property of the vote, not of the compressor (Bernstein et al. 2019). In the error-feedback variants the LMO moves from the clients to the server: each client sign-compresses the residual between its local momentum and its gradient estimator and sends the pair $(\mathbf{s}_t^{(j)}, \alpha_t^{(j)})$; the server aggregates these, updates a global gradient estimator, and applies the Muon LMO. The extra per-layer scalar keeps the uplink at ≈ 1 bit per parameter, but the server-side quantity is now dense — a scaled average of signs, then the real-valued $\text{polar}(\cdot)$ of it — so the vote argument does not apply and the full model must be broadcast, unless a second error-feedback loop compresses it, which is what EF21-MuonSign adds. The complete procedures — gradient accumulation, the EMA momentum, and the AdamW treatment of the classifier head — are Algorithms A8–A9 (Appendix A.18).

5 Experiments

A controlled study on a deterministic smooth convex quadratic — measuring the gradient/step alignment that Theorems 1–3 drive negative, and separating the accuracy floor of a sign step from its rate — is in Appendix A.11.

Method	η_0	Test acc (%)	Ep. to 90%
EF21-SignMuon	0.01	94.64 ± 0.24	8.7
SignMuon	0.02	94.53 ± 0.22	8.3
Muon	0.05	94.49 ± 0.09	7.7
EF21-MuonSign	0.01	94.22 ± 0.13	10.3
EF21-MuonUSign	0.2	94.15 ± 0.16	8.7
MuonUSign	0.02	93.90 ± 0.08	10.0
MuonSign	0.005	93.65 ± 0.21	15.0
Adam	0.0005	93.35 ± 0.15	18.8
SignSGD	0.002	93.27 ± 0.16	19.0
SGD	0.02	93.13 ± 0.07	21.7

Table 1: Centralized ResNet-18 / CIFAR-10, 75 epochs. Test accuracy is the mean $\pm$ standard deviation over three seeds of the last five epochs; “Ep. to 90%” is the mean epoch at which test accuracy first reaches 90%. Rules separate the statistically tied methods from the rest.

5.1 Centralized Learning

We compare all six matrix-aware 1-bit methods against Muon, SignSGD, SGD and Adam at a fixed epoch budget, on CIFAR-10 (Krizhevsky, Hinton et al. 2009). The base model is a ResNet-18 adapted to low-resolution images: an initial 3×3 convolution without max-pooling, 4 residual blocks with BatchNorm, dropout, and a linear classifier. Training runs for $T_{\max} = 75$ epochs with η_0 annealed to zero by a cosine schedule — the convention of the domain rather than the constant rate our rates are proved for (Appendix A.10).

Protocol. The only tuned hyperparameter is η_0 , selected per method on a held-out 5k validation split; **the test set is never consulted during tuning**. Every method with a norm-fixed step is tuned and reported under one per-layer rule, the unit-gain rule (9), so that η_0 means the same thing — the per-step RMS gain — across them (Appendix A.17); SGD and Adam have no such step and keep one global rate. Centralized numbers are means over three seeds and federated numbers over five, and differences smaller than the seed spread are not claimed as results. Alongside final accuracy we report epochs to a fixed test accuracy, which separates the methods by factors rather than by tenths of a point. The grid, the endpoint rule and the schedules are in Appendix A.10.

EF21-SignMuon, SignMuon and Muon span 0.15 points against seed spreads of 0.09–0.24; at three seeds they are indistinguishable, so we claim only that sign-after *matches* Muon — at one bit per parameter on the uplink, and 1.26 points above SignSGD, which spends the same budget without the LMO. Error feedback is free — EF21-SignMuon sits within a seed spread of uncompressed SignMuon.

Within the family the ordering exceeds the noise: sign-after (94.53), sign-before (93.90) and bidirectional (93.65) separate by 0.9 points, and the training loss orders them identically (Figure A6), so the gap is in optimization rather than regularization. SignMuon crosses 90% in 8.3 epochs against SGD’s 21.7, SignSGD’s 19.0 and Adam’s 18.8; orthogonalization costs about 18% more per epoch (14.4 vs. 12.2 s, one device throughout), so $2.6\times$ fewer epochs is still $2.2\times$ in wall clock.

Optimizer	η_0	Up	Down	Rds. to 80%	Test acc (%)
Muon	0.1	32	32	260	85.99 ± 0.26
SignMuon	0.05	1.09	1.09	300	85.81 ± 0.25
EF21-SignMuon	0.05	1.09	32	600	85.16 ± 0.15
MuonUSign	0.05	1.09	32	540	84.62 ± 0.17
EF21-MuonSign	0.05	1.09	1.09	980	83.99 ± 0.28
EF21-MuonUSign	0.05	1.09	32	1000	83.75 ± 0.21
SignSGD	0.02	1.09	1.09	1080	81.83 ± 0.43
MuonSign	0.05	1.09	1.09	1300 [†]	80.68 ± 1.88
SGD	0.1	32	32	1367 [†]	79.79 ± 1.55
Adam	0.001	32	32	—	77.12 ± 1.04

Table 2: CIFAR-10 federated learning on CNN2, ordered by accuracy: $N = 11$ clients, homogeneous split, 2000 rounds of 3 local steps at batch 64, momentum 0.9. Test accuracy is the mean of the final five evaluations over five seeds, $\pm$ one standard deviation *across* seeds. **Up** and **Down** are mean bits per model parameter per round in each direction (Appendix A.14). [†] mean over the seeds reaching 80% within the budget (4/5 for MuonSign, 3/5 for SGD); Adam reached it on none.

5.2 Federated Learning

In federated learning the bottleneck is transmitting full 32-bit gradient matrices. We compare all six matrix-aware 1-bit methods against the Muon, SignSGD, SGD and Adam baselines on CNN2 — two convolutional blocks with BatchNorm and an MLP classifier — with CIFAR-10 partitioned homogeneously across clients. Local models are discarded at the end of each round, so BatchNorm runs at its initialization statistics throughout, in training and evaluation alike (Appendix A.10). Two schemes are covered: the worker-side-LMO scheme (Algorithm A8), of which federated SignMuon is the instance, and the server-side-LMO scheme (Algorithm A9), which covers the error-feedback variants; Table A12 lists which compressor pair fixes each method.

Setting. Sign-based methods separate most clearly as network scale and local computation grow, so we run at $N = 11$ clients with 3 local steps each, the scale at which they separate. The count is odd by design: client messages are ± 1 valued under the convention of Section 4, so an odd N makes the majority vote decisive in every coordinate and removes the tie-breaking rule from the comparison entirely. Every rate is tuned per method on a validation split held out before the client partition, so that no placement is handicapped by its step size and no test image is scored during selection (Appendix A.10); every number is a five-seed mean (Table 2, Figure A8).

SignMuon reaches the quality of Muon: $85.81 \pm 0.25\%$ against $85.99 \pm 0.26\%$, a gap of 0.18 points against a seed spread of 0.25, at 1.09 bits per parameter in *both* directions. That is the cleanest statement the experiment supports: at a fixed one-bit round trip, replacing the gradient sign with the sign of the LMO direction is worth 3.98 points over SignSGD — against 1.26 centrally — and closes

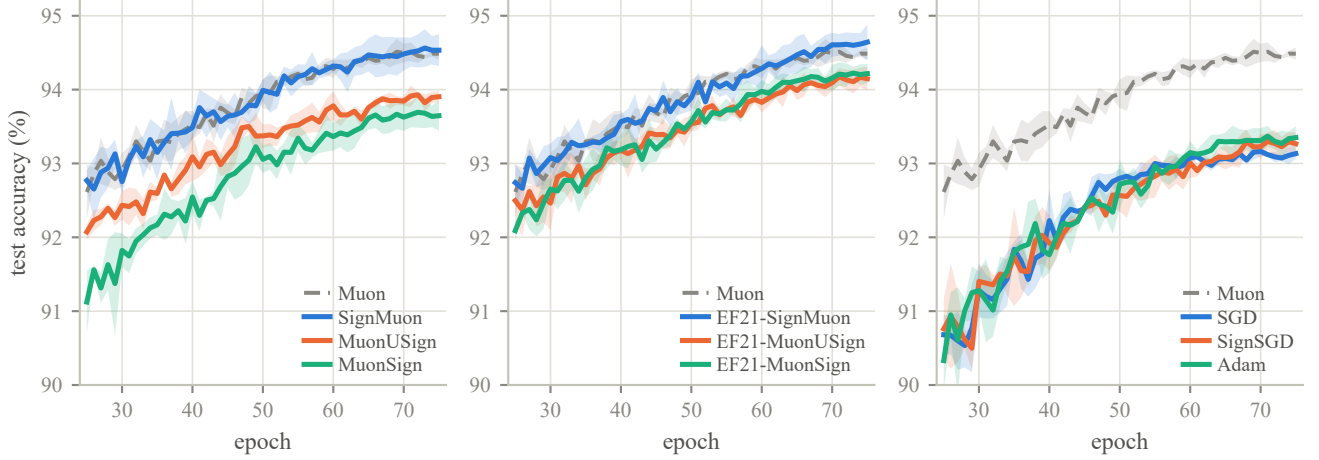


Figure 2: Centralized ResNet-18 on CIFAR-10: test accuracy from epoch 25, at each method’s selected η_0 (Table 1). Panels group the three sign placements (left), the EF21 variants (centre) and the uncompressed baselines (right); Muon is the gray dashed reference in all three, and bands are ± 1 standard deviation over three seeds.

the gap to uncompressed Muon to within the seed spread. The seeds also make visible that the three placements are *not* interchangeable: sign-after leads (SignMuon, 85.81%), sign-before follows 1.19 points back (MuonUSign, 84.62%), and signing both sides costs a further 3.94 (MuonSign, 80.68%). Sign-after stays ahead under error feedback (EF21-SignMuon, 85.16%), though below it the other two swap (EF21-MuonSign 83.99%, EF21-MuonUSign 83.75%).

This is the paper’s uncomfortable finding: the placement that works best in practice is the one the theory condemns. Sign-after is what Theorem 1 makes diverge, and error feedback does not repair it for any (L, η, μ) (Theorem 4); yet the sign-after methods are the strongest compressed ones here, in the centralized table, and on nanoGPT (Section 5.3). Conversely the two variants that do carry an unconditional $\mathcal{O}(T^{-1/2})$ guarantee, EF21-MuonUSign and EF21-MuonSign, sit about 2 points below SignMuon — seven to eight seed spreads, so the gap is measured rather than inferred. We therefore state the guarantee as costing accuracy at this scale rather than being free, and read the counterexamples as what they are: statements about worst-case instances, not predictions about CIFAR-10.

EF21-MuonSign is 0.24 points *ahead* of the uplink-only EF21-MuonUSign, within a seed spread, so at CNN2’s scale the one-bit downlink loop is free: what separates the EF21 family from SignMuon is the sign placement, not the second compression. All methods here are evaluated on the exact server model $\mathbf{X}_t$, the iterate Corollary 1 bounds, which for EF21-MuonSign is not the model the clients hold (17); that the two agree at CNN2’s scale means the models still track each other. On much wider matrices they part (Remark 3), and the next section reports both.

5.3 Language Modelling

We test the methods where the matrices are large enough for the layer-rank dependence of Corollary 1 to bite: the modded-

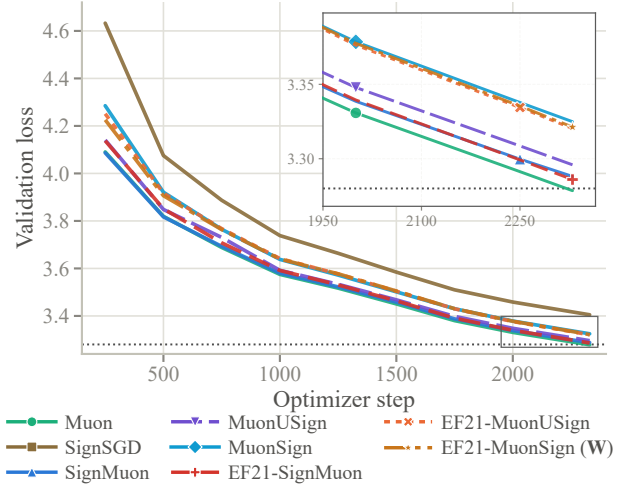


Figure 3: NanoGPT speedrun, $8 \times H100$: validation loss against optimizer step, single seed; dotted is the speedrun target 3.28. EF21-MuonSign is drawn at its broadcast model **W**. The inset magnifies the boxed tail, where SignSGD is above the range shown; exact values are in Table 3.

nanoGPT speedrun (record #40), a 12-layer transformer of model dimension 768 with 768×3072 hidden matrices, trained on 611M FineWeb tokens on $8 \times H100$. Each method takes over the record’s Muon on the hidden matrices and gates; embeddings, scalars and the head keep the record’s Adam. Our Muon arm *is* record #40’s own optimizer and reproduces its published result to 0.6 sd, which validates the port before anything else is read. Rates are not tuned per method but fixed a priori by the unit-gain rule at one η_0 per *family* – 0.06 for the LMO-terminated methods, the record’s own value, and 0.03 for the sign-terminated ones – so every

Method	Step	η_0	Val. loss	Steps to 3.35
Muon (record #40)	LMO	0.06	3.2785	1.00×
EF21-SignMuon	LMO	0.06	3.2860	1.02×
SignMuon	SIGN	0.03	3.2881	1.02×
MuonUSign	LMO	0.06	3.2959	1.05×
EF21-MuonUSign	LMO	0.06	3.3203	1.13×
EF21-MuonSign (W)	LMO	0.06	3.3213	1.14×
MuonSign	SIGN	0.03	3.3249	1.14×
SignSGD	SIGN	0.03	3.4049	—
EF21-MuonSign (X)	LMO	0.06	5.5198	—

Table 3: NanoGPT after 2330 steps (611M tokens), single seed. Record #40 reports 3.2780 ± 0.0009 over five seeds, so differences below ~ 0.003 are noise. “Steps to 3.35” is the step count to reach that loss, relative to Muon. The last row is EF21-MuonSign’s server model **X**, the same run as the broadcast model **W** above.

contrast is matched-hyperparameter, at the cost of not reporting each method’s best achievable loss. Runs are single-seed, and Table 3 states the seed noise of the reference so that the resolvable differences are explicit (Appendices A.16 and A.17).

Three things follow (Table 3, Figure 3). First, the matrix-aware 1-bit methods transfer to language modelling: all of them beat SignSGD by 0.08–0.12 nats, and the best of them sit within 0.01 of full-precision Muon, at equal wall-clock. Second, error feedback is cheap but not free. Its clean cost is the matched pair $\text{MuonUSign} \rightarrow \text{EF21-MuonUSign}$, +0.024 nats or $\sim 8\%$ more steps — the price of steering the LMO with an estimator that lags its target. EF21-SignMuon, whose target $\text{polar}(\tilde{\mathbf{M}}_t)$ is orthogonal and hence isotropic, pays almost nothing and is the strongest compressed method here, notwithstanding Theorem 4.

Third, the bidirectional method’s two models (17) come apart here. Its broadcast model **W** — the one every gradient is evaluated at — is indistinguishable from EF21-MuonUSign (3.3213 vs 3.3203), so the compressed *downlink* costs nothing where training happens. The server model **X**, which is what Corollary 1 bounds, settles ≈ 2.2 nats above it: a persistent offset, not a divergence, and one that Appendix A.16 localizes to the single zero-initialized c_proj layer. This is the mechanism of Remark 3 rather than a tuning failure, and lowering η_0 does not repair it — the admissible step would have to shrink by a further $\sqrt{r}$ in the layer rank (Remark 4) — but a spectrally contractive downlink compressor would.

6 Conclusion

SignMuon applies 1-bit sign compression to the Muon LMO direction: it matches Muon’s accuracy and outperforms SignSGD, but we prove that a directly sign-compressed LMO has no global convergence guarantee and can diverge. EF21-MuonUSign compresses the internal residual instead, which restores a transferred $\mathcal{O}(T^{-1/2})$ guarantee and descent on our counterexamples at a 1-bit uplink, and lands about two accuracy points below uncompressed SignMuon;

EF21-MuonSign brings the downlink to one bit at no measured cost.

An open tension remains. On every task we measure — centralized, federated and language modelling — the strongest compressed method places the sign after the LMO, exactly the placement Theorems 1 and 4 show to be unrepairable in the worst case. The practitioner chooses between a bound that holds on every instance and roughly two accuracy points on the instances they have; narrowing that gap, by finding what property of realistic problems the counterexamples violate, is the natural continuation of this work.

Future work. Our (L^0, L^1) guarantee (Corollary 2) covers the uplink-compressed EF21-MuonUSign only, since the generalized-smooth theory of Gruntkowska et al. (2025) is available only without downlink compression; extending it to EF21-MuonSign is open. So is the downlink itself: the rank penalty EF21-MuonSign pays comes from compressing the model in the wrong norm, removable only by changing the compressor (Remark 4), and whether *any* one-bit compressor can be layer-norm contractive we do not know. Since Gruntkowska et al. (2025) fix the server compressor to the identity throughout their experiments, this regime appears not to have been probed empirically before. Finally, the reduction of Appendix A.9 applies verbatim to arbitrary layer geometries (Remark 5), and how these methods scale to LLM-sized matrices remains to be measured.

References

- Ahn, K.; Xu, B.; Abreu, N.; Fan, Y.; Magakyan, G.; Sharma, P.; Zhan, Z.; and Langford, J. 2025. Dion: Distributed Orthonormalized Updates. arXiv:2504.05295.
- Bernstein, J.; and Newhouse, L. 2024. Old Optimizer, New Norm: An Anthology. arXiv:2409.20325.
- Bernstein, J.; Wang, Y.-X.; Azizzadenesheli, K.; and Anandkumar, A. 2018. signSGD: Compressed Optimisation for Non-Convex Problems. arXiv:1802.04434.
- Bernstein, J.; Zhao, J.; Azizzadenesheli, K.; and Anandkumar, A. 2019. signSGD with Majority Vote is Communication Efficient And Fault Tolerant. arXiv:1810.05291.
- Beznosikov, A.; Horváth, S.; Richtárik, P.; and Safaryan, M. 2024. On Biased Compression for Distributed Learning. arXiv:2002.12410.
- Bolatov, A.; Riabinin, A.; Kornilov, N.; Veprikov, A.; Horváth, S.; Takáč, M.; and Beznosikov, A. 2026. Lion-Muon: Alternating Spectral and Sign Descent for Efficient Training. arXiv:2605.19811.
- Chen, X.; Chen, T.; Sun, H.; Wu, Z. S.; and Hong, M. 2020. Distributed training with heterogeneous data: bridging median- and mean-based algorithms. In *Proceedings of the 34th International Conference on Neural Information Processing Systems*, NIPS ’20. Red Hook, NY, USA: Curran Associates Inc. ISBN 9781713829546.
- Essential AI. 2025. Practical Efficiency of Muon for Pre-training. arXiv:2505.02222.
- Gruntkowska, K.; Gaponov, A.; Tovmasyan, Z.; and Richtárik, P. 2025. Error Feedback for Muon and Friends. arXiv:2510.00643.

Horváth, S.; Kovalev, D.; Mishchenko, K.; Stich, S.; and Richtárik, P. 2019. Stochastic Distributed Learning with Gradient Quantization and Variance Reduction. arXiv:1904.05115.

Jin, R.; Liu, Y.; Huang, Y.; He, X.; Wu, T.; and Dai, H. 2025. Sign-Based Gradient Descent With Heterogeneous Data: Convergence and Byzantine Resilience. *IEEE Transactions on Neural Networks and Learning Systems*, 36(2): 3834–3846.

Jordan, K.; Jin, Y.; Boza, V.; You, J.; Cesista, F.; Newhouse, L.; and Bernstein, J. 2024. Muon: An optimizer for hidden layers in neural networks.

Karimireddy, S. P.; Rebjock, Q.; Stich, S.; and Jaggi, M. 2019. Error Feedback Fixes SignSGD and other Gradient Compression Schemes. In Chaudhuri, K.; and Salakhutdinov, R., eds., *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, 3252–3261. PMLR.

Kravatskiy, A.; Kozyrev, I.; Kozlov, N.; Vinogradov, A.; Merkulov, D.; and Oseledets, I. 2025. Ky Fan Norms and Beyond: Dual Norms and Combinations for Matrix Optimization. arXiv:2512.09678.

Krizhevsky, A.; Hinton, G.; et al. 2009. Learning multiple layers of features from tiny images.

Large, T.; Liu, Y.; Huh, M.; Bahng, H.; Isola, P.; and Bernstein, J. 2024. Scalable Optimization in the Modular Norm. arXiv:2405.14813.

Li, J.; and Hong, M. 2025. A Note on the Convergence of Muon. arXiv:2502.02900.

Mishra, N.; Trivedi, K.; and Kumar, P. 2026. SignMuon: Communication-Efficient Distributed Muon Optimization. arXiv:2605.16311.

Qian, X.; Gaponov, A.; Malinovsky, G.; and Richtárik, P. 2026. Communication-Efficient Gluon in Federated Learning. arXiv:2604.10689.

Riabini, A.; Shulgin, E.; Gruntkowska, K.; and Richtárik, P. 2025. Gluon: Making Muon & Scion Great Again! (Bridging Theory and Practice of LMO-based Optimizers for LLMs). arXiv:2505.13416.

Richtárik, P.; Sokolov, I.; and Fatkhullin, I. 2021. EF21: A new, simpler, theoretically better, and practically faster error feedback. In *Advances in Neural Information Processing Systems*, volume 34, 4384–4396.

Safaryan, M.; and Richtarik, P. 2021. Stochastic Sign Descent Methods: New Algorithms and Better Theory. In Meila, M.; and Zhang, T., eds., *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, 9224–9234. PMLR.

Yang, G.; Simon, J. B.; and Bernstein, J. 2023. A Spectral Condition for Feature Learning. arXiv:2310.17813.